# Chaotic Circuit [1]

## Objectives

- To understand how an op amp functions and how to use it in a circuit
- To explore a chaotic system
- To use numerical simulations to guide and enhance experimental work

## Background Reading

### Primary

- K. Kiers, D. Schmidt and J. C. Sprott, "Precision measurements of a simple chaotic circuit", *Am. J. Phys.* **72**, 503–509 (2004). (On Canvas)
- S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd Ed., 2015, reprinted by CRC Press, 2018. (Can be downloaded from SFU Library.) Has the best overview of chaos and nonlinear dynamics. If nothing else, read the first chapter. Chapter 10 has good insight on bifurcation diagrams and discusses other ways to characterize chaotic behaviour.
- Advanced Physics Laboratory Handout *Electronics Tips*. (On Canvas)

### Supplemental

- P. Horowitz and W. Hill, *The Art of Electronics*, Cambridge University Press. (Chapter 4: "Feedback and Operational Amplifiers" is on Canvas). More info on op-amps.
- G. L. Baker and J. P. Gollub, *Chaotic Dynamics: an Introduction*, Press Syndicate of the University of Cambridge, 1996. (In SFU Library.)
- J. R. Taylor, *Classical Mechanics*, University Science Books, 2005. See Chapter 12, "Nonlinear Mechanics and Chaos". (On Canvas)

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# 1 Introduction

In previous physics classes, you have studied classical and some quantum mechanics, electricity and magnetism, waves, and the like. All these subjects share the common feature of having equations of motion, from Newton's laws of motion to AC circuit equations, to equations for heating and cooling. You have also learned to solve these equations for many particular examples.

One can take a more general point of view and ask, What kinds of behaviour are possible for different kinds of equations of motion? This is the question addressed in the mathematical subject of *dynamical systems theory*. The properties of differential equations are part of this subject, but it also includes discrete equations and more. Focusing on continuous-time dynamics, systems with friction (or *dissipation*) tend to slow and stop, unless they are *driven*. If they are continuously driven, they can settle in to several types of behaviour: steady state, periodic motion, and even an irregular motion known as *chaos*. Most of the interesting features arise when the equations of motion are *nonlinear*. These are much harder to solve, and a variety of approaches, ranging from numerical simulation to geometric diagrams to analytic approximation have been developed for their study. The Strogatz text listed above is a good introduction to the field and these techniques.

In this experiment, you will build an electrical circuit that exhibits chaotic behaviour and study phase portraits and bifurcation diagrams of the system through both experiment and simulation.

The circuit models a particular example of a class of nonlinear dynamical systems that exhibit chaotic behaviour. This class of examples is described by an ordinary differential equation in a single variable that is third order and contains a nonlinearity,

$$\dddot{x} = -A\ddot{x} - \dot{x} + D(x) - \alpha\,, \tag{1}$$

where $A$ and $\alpha$ are constants, $D(x)$ is a nonlinear function, and the dots represent derivatives with respect to a dimensionless time. Since the third derivative of the position is also known as the *jerk*, an equation of this form is also called a *jerk equation* and circuits that model jerk equations are known as *jerk circuits*.[2]

The circuit, shown in Fig. 1(a), is described in detail in the main reference article by Kiers et al. It consists of a series of three successive inverting integrators, with outputs at $V_2 = \ddot{x}$, $V_1 = -\dot{x}$ and $x$, a summing amplifier with output at $V_3$, and the nonlinear element $D(x)$. A schematic for the nonlinear element is shown in Fig. 1(b). Its two parameters, $A$ and $\alpha$, play different roles. The latter, $\alpha$, is a *system parameter*. Its value depends on the values of the fixed resistors and capacitors used in the circuit and is fixed for all experiments. The former, $A$, is a *control parameter* and can be systematically varied. You will see that $A = R/R_v$, with $R_v$ a variable resistor and $R$ a fixed resistor.

When $R_v$ varies, so does $A$, and the dynamical behaviour of the voltage will also vary. For $A > 1$, the system oscillates with a period that varies smoothly with $A$. As $A$ is decreased (or $R_v$ increased), there is a special value $A_1$ where the period all of a sudden changes from $T$ to $2T$ by having every other oscillation have a slightly different amplitude. This change in period is known

[2] Users of such circuits are <u>not</u> known as *jerks*....

as *period doubling*. As $A$ is further decreased the period again varies slightly until at another special value $A_2$, the period goes from $2T'$ to $4T'$. There has been a second period doubling. The abrupt (non-analytic) change in behaviour at a special value of $A$ is called a *bifurcation*.

As the control parameter is further decreased, there is a sequence of period-doubling bifurcations. The bifurcation values of $A$ become closer and closer together, eventually accumulating at a value $A^*$ where the period $\sim 2^n$ with $n \to \infty$ becomes infinite. At this point, the period is infinite, and motion never repeats: the system is now *chaotic*. Beyond, the motion alternates between other periodic windows (e.g., period 3), with their own period-doubling cascades. All this complexity from such a simple circuit!

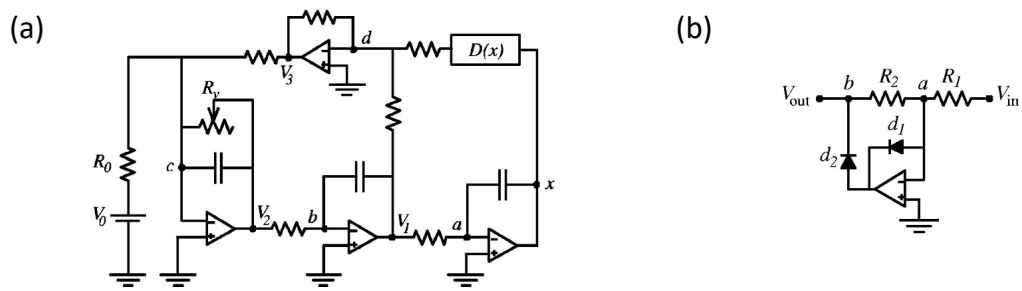



**Figure 1:** Schematic diagram of the circuit to be tested in this experiment: (a) main circuit, (b) nonlinear element. Nominal values for the unlabeled resistors and capacitors are $R = 47$ k$\Omega$ and $C = 1$ nF. Approximate values for the input voltage and resistor are $V_o = 0.300$ V and $R_o = 157$ k$\Omega$. Values for $R_1$ and $R_2$ should be chosen of order 10's of k$\Omega$ (important) and so that the ratio is $R_2/R_1 \approx 6$ (i.e. approximately 6). (Figure reproduced from the article by Kiers et al.)

The experiment has two parts: The first part introduces basic properties of op amps so that you can test and understand parts of the overall circuit; the second part focusses on putting together those parts into a single circuit and exploring its dynamics.

# 2 Part I—Op amps

The jerk circuit is made from several sub-circuits, each with a simple function, such as summing two signals, amplifying a signal (multiplying by a constant), and integrating. The common feature of the sub-circuits is that they all use an electrical element known as an *operational amplifier*, or "op amp" to its friends.

An op amp, is a high-gain differential amplifier. It is itself an electrical circuit made of transistors and the like, but it is almost always encountered as an integrated circuit ("chip") that has a few simple properties: There are two inputs $V_{\text{in},+}$ and $V_{\text{in},-}$, known as the non-inverting and the inverting input, respectively, and one output, $V_{\text{out}}$. Characteristics include

- High gain, on the order of $10^6$

- High input impedance, low output impedance
- Used with a $\pm$ 12 or $\pm$ 15 V supply
- Relatively insensitive to temperature and power supply fluctuations
- Used with feedback

The standard representation is a sideways triangle with inputs and output, as shown in Fig. 2a. The hardware itself often takes the form of an 8-pin "dual inline package", or DIP, shown in Fig. 2b. The pins of a DIP are numbered sequentially, counterclockwise. Note the half-moon cut out on one end: Pin 1 is located to the left of the cut out and Pin 8 to the right. Conventionally, the DIP is always oriented on a breadboard so that the cut out is on the left.

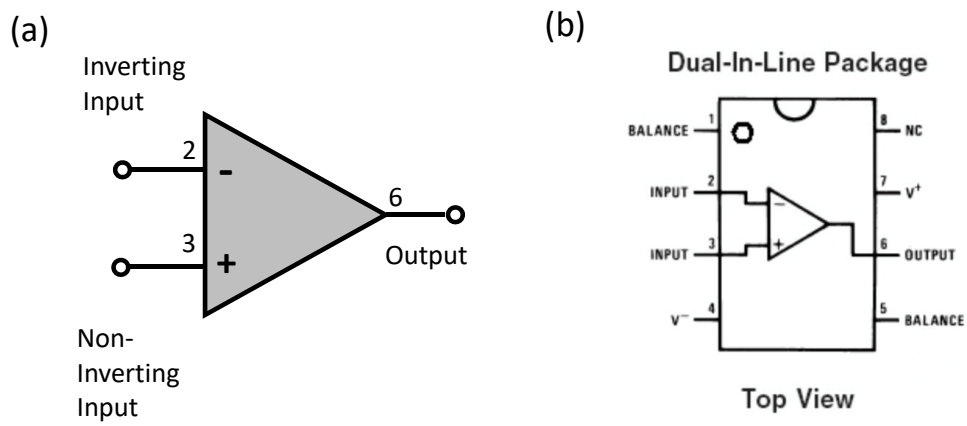



**Figure 2:** Op amp diagrams: (a) standard circuit diagram representation; (b) typical connections. Connect $V^+$ and $V^-$ to the +15 and −15 V terminals on your power supply. Note that the connection to the power leads is assumed and is not usually shown in the circuit diagram. Also note that "NC" stands for no connection. The two "BALANCE" pins, which are used to zero the outputs of the opamp, will not be used in this lab. The three NC and BALANCE pins should be left floating. The other five should have connections. Be sure to verify the pin configuration for *your* op amp.

When used with negative feedback, ideal op-amp circuits approximately obey two simple rules:

### The op amp Golden Rules

1. The inputs draw no current: The input impedance is so high that it is effectively infinite.
2. The output attempts to do whatever is necessary to make the voltage difference between the two inputs zero; i.e., $V_{\text{in},+} = V_{\text{in},-}$.

In Rule 2, the word "attempts" is used because the output voltage of the op-amp cannot exceed the supply voltages and will "clip", or limit, the voltage when it gets too close to those bounds.

### Example: The voltage follower

The simplest op-amp circuit is the follower, shown in Fig. 3. The first golden rule implies that the

input impedance is infinite. The second rule is more interesting: Since the inverting input ($V_{\text{in},-}$) is wired to the output, these two voltages must be equal. The only way that the two inputs can be at the same voltage, $V_{\text{in},-} = V_{\text{in},+}$, is for $V_{\text{out}} = V_{\text{in}}$. Consequently the op amp behaves like a perfect voltage follower. (Why build such a circuit? Many signal sources are unable to maintain a voltage if much current is drawn. Op amps can typically maintain the desired voltage *and* source higher currents.)

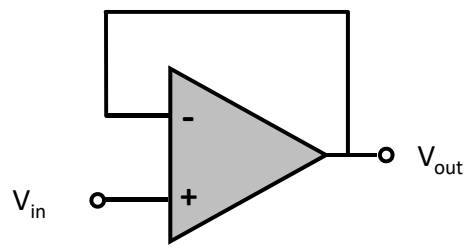



**Figure 3:** Op-amp follower: $V_{\text{in}}$ is connected to the non-inverting input, and the inverting input is connected to the output.

## Experiments

The goal of these initial experiments is to test some simple sub-circuits of the more complicated nonlinear circuit that we will explore later. You will also gain experience working with op amps and can examine the extent to which their real behaviour is well modeled by the golden rules.

Build the circuits described below. Describe your procedure and observations in your lab notebook. Include a circuit diagram, including power connections, input and output voltages. Use the function generator to produce an AC input signal. The function generator can also produce a simple DC voltage by pushing and holding the DC offset button. Experiment with different types of op amps to see how they behave. Include an analysis of the circuit based on the two golden rules and verify that the circuits behave as expected. In each, make sure you understand how $V_{\text{out}}(t)$ is related to $V_{\text{in}}(t)$—as predicted and observed.

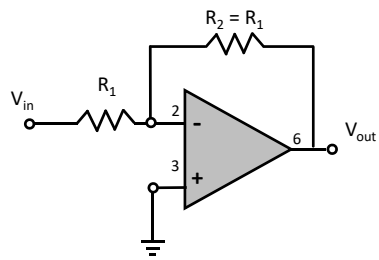
### Equipment

- Breadboard with built-in power supplies (±15 V and +5 V).
- Function generator (Keysight 33210A or 33120A).
- Siglent oscilloscope (SDS2352X-E) and oscilloscope probes.
- LF411 op amps in drawers at the back of the room — be sure to look up the pin configuration ("pinout") of the op amp.
- Diodes (1N4148 or 1N914).
- Resistors, capacitors also available at the back of the room.

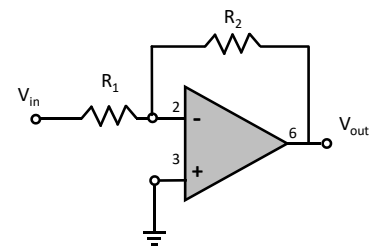
- Selection of cables and adapters. Hook up wires, hook-up wire, additional cables and connectors at back of room.
- Variable DC power supply, or if working with the Keysight 33210A function generator use the DC Volts feature (Utility ¿ DC on)

If your circuit is not behaving as expected, consider testing your op amp to make sure that it is working properly. Be sure to use the provided high-impedance oscilloscope probes to monitor voltages. Also, make sure the input ($\times 1$ or $\times 10$) is set correctly. See the note in Part 3 in the main part of the experiment.

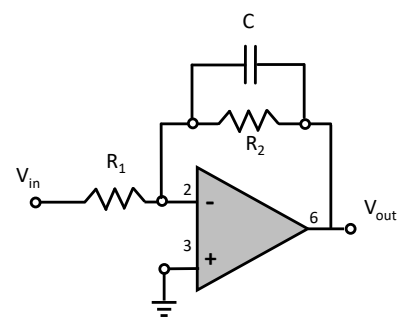
**1. Unity-gain, inverting amplifier.** Note the two resistors have the same resistance. $R_2$ is the *feedback resistor*. Use $R_1 = R_2 = 10$ k$\Omega$. Try a 1-kHz triangle wave with zero offset (to provide a range of input voltages) as an input signal and use an oscilloscope to measure the input and output voltages. To measure the input voltage on the oscilloscope, use a T-adapter and BNC cable to split the signal from the function generator.



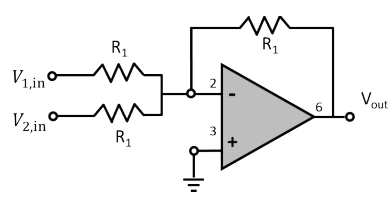

**2. Inverting amplifier.** Show that the gain of the amplifier is $-R_2/R_1$. The gain is defined as $V_{\text{out}}/V_{\text{in}}$. The sign change corresponds to a 180° phase shift.




**3. Practical integrator.** A practical integrator, shown in the figure on the right, is a basic integrator modified by the addition of a large-resistance feedback resistor $R_2$. Use a function generator for the input (for example a square wave) and an oscilloscope to measure the input and output voltages. Again use the two golden rules to predict the output. This time you will to write down some time-dependent equations for both halves of the input cycle. Try $R_1 = 10$ k$\Omega$, $R_2 = 100$ k$\Omega$, and $C_1 = 1$ nF. The purpose of $R_2$ is bleed off charge buildup in the capacitor that may occur in the presence of a small DC voltage. Can you predict the form of $V_{\text{out}}(t)$ when $V_{\text{in}}(t)$ is a square wave? With $R_2$ removed, try to vary the DC offset on the function generator to see what happens to the output.




**4. Summing amplifier.** $V_{\text{out}} = -\left(V_{\text{in},1} + V_{\text{in},2}\right)$. For one input signal you can use the function generator. For the second input, you can use a DC voltage from a variable power supply, or you can create your own variable DC voltage from one of the output voltages on your breadboard together with a potentiometer. [3]



**Pre-lab Question 1** (to be completed by the end of Day 1): The practical integrator circuit (#3, above) is not exactly an integrator.

i. Derive the differential equation relating $V_{\text{out}}(t)$ to $V_{\text{in}}(t)$.
ii. Solve this equation for $V_{\text{out}}(t)$ as a function of $V_{\text{in}}(t)$. You should find that there is an integral involved, but there is a twist, related to the presence of $R_2$.
iii. For $V_{\text{in}}(t)$ a square wave with period $T$ and zero mean, plot schematic solutions for $T \gg R_2C$, $T \approx R_2C$, and $T \ll R_2C$. For which of these three cases is the output most like a simple integral? (Note: consider approximate forms of the expression derived in (ii) in the limit of small and large $R_2C$ time constant to understand the two extremal cases.)
iv. What problem for the circuit can $R_2$ help prevent?

# 3 Part II—Chaotic Circuit

**Pre-lab Question 2** (to be completed before you start Part II):

Apply what you learned in Part I to the circuit in Fig. 1 to derive Eqs. 2–6 in the paper by Kiers et al.

### Equipment

- breadboard
- function generator
- oscilloscope
- op amps (LF411), diodes (1N914 or 1N4148)[4], resistors, capacitors, cables are available at the back of the room
- decade resistor box (manual stepping resistor)
- potentiometer (variable resistor)
- oscilloscope for data acquisition

Here is a list that breaks down some of the tasks required to build and model the chaotic circuit.

1. Build and test the nonlinear element $D(x)$. Values for $R_1$ and $R_2$ should be chosen of order 10's of k$\Omega$ (important) and so that the ratio is $R_2/R_1 \approx 6$ (i.e. just approximately 6).

[4]These are fast signal diodes with low reverse current, like BAV20 as used in Kiers et al. A lower performance general purpose diode like N4001 can be used in a pinch for a qualitatively working circuit but aren't suitable for precision work.

2. For testing the circuit, use a 1-kHz sawtooth wave as input and record both input and output.
3. Build and test each op amp separately. We suggest that you use 1 nF capacitors for the three integrators, rather than the 1 µF capacitors used by Kiers et al. This will speed up the circuit greatly[5] but not change its qualitative behaviour. Read Kiers et al. to understand why this is so.
4. To test each op amp element, add a temporary 500-kΩ feedback resistor in parallel with the capacitor and test the output for an AC input (e.g., square wave—choose its frequency carefully, following Prelab Q1). Note: If you do not add the resistor, the capacitor can charge up due to any small input dc offset and the op amp will probably rail (i.e., the output voltage will increase to either $V^+$ or $V^-$). Op amps are easily destroyed—it is probably worth repeating this with each op amp each week. If your op amp is not working as expected, throw it out.
5. After testing, remove the temporary large feedback resistors from the integrators, and add the dc voltage offset $V_0$ to the circuit (from a DC power supply or function generator in DC voltage mode).
6. Set the circuit to initial conditions in which you expect a single stable solution. Vary $R_v$ to change the dynamics of the circuit. We supply a resistor box that can make both coarse and fine changes in $R_v$.
7. Create the bifurcation diagram of the system, record sample phase portraits and calculate power spectral densities. See below for suggestions of more advanced features to look at.
8. Analyze the circuit numerically and compare to the data you generate with the actual circuit. For example, you should be able to simulate waveforms for various values of $R_v$, prepare plots of $x(t)$ as a function of time, and the phase portrait plot of $\dot{x}$ as a function of $x$. Note that we have provided a Python demo notebook to help get you started solving ordinary differential equations numerically[6] . See `ODE Solving.ipynb`.

**Pre-lab Question 3** (to be completed before starting the fourth Lab Period):

Numerically solve the ODE that your circuit is designed to represent. (Hint: write the third-order equation as three coupled first-order equations.) Find solutions in the region of parameter space where you expect to find a single periodic solution (stable fixed point), after it bifurcates and has two periodic solutions, and in the region where you expect chaotic behaviour.

Determine the maximum amplitudes of the different solutions.

[5] It makes it 1000× faster and 10× more fun!

[6] Note the demo notebook will get you started but as always it's not "plug-n-play." You need to check how the scipy integrator works and optimize its performance to get reliable output.

**Use your circuit to investigate**

- Period doubling
- Bifurcation plots
- Phase portraits
- Power density plots

**Things to try if you have time**

- Set up an audio amplifier to "hear" chaos. For this, you may need to swap the 1 nF capacitors for larger values to give characteristic frequencies in the audio range (try 10 nF).

# 4 SUGGESTED TIMELINE

Lab Period 1:

- Aim to complete most of Part I today—this is meant to introduce you to the use of op amps

Lab Period 2:

- Hand in Pre-lab Question 1 before you start Part II
- Set up and test the nonlinear element that characterizes the nonlinearity of your circuit

Lab Period 3:

- Build your circuit
- Observe chaos

Lab Period 4:

- Hand in Pre-lab Question 2
- Take some data to compare to your simulation

Lab Period 5:

- Take better data; analyze and compare to simulation
- Make phase portraits (for example, using the two oscilloscope channels), bifurcation diagrams and power spectrum plots.
- Try for more sophisticated measurements (Feigenbaum $\alpha$, return map, ....)

Lab Period 6:

- Complete your data set and analysis.

Lab Period 7-8:

- Additional time for data collection, analysis and projects.

# 5 Items to include in your documentation

In addition to a record of day-to-day happenings, calculations, diagrams, essential equations, lists of equipment, filenames, etc. please make sure you include at least the following in either your lab notebook or your analysis notebook.

**Figures**

- circuit diagrams
- nonlinear element test
- raw data for signal for different conditions—simulated and measured
- maxima plot—simulated and measured
- phase portraits—simulated and measured
- bifurcation diagram—simulated and measured. For this, you will need a way to estimate peak positions in the oscillating voltage $x$ signal. One way is to use zero crossings of its derivative $\dot{x}$. Alternatively, you can try to identify each peak approximately and then fit to a local quadratic maximum about it. You should check whether your method is accurate (look at the positions it finds...). Also, you will need to deal with the effects of noise, which could lead to spurious multiple "peaks". You can adjust the tolerance in the ODE solver to improve th quality of the solution.

**Analysis Notes and additional project ideas**

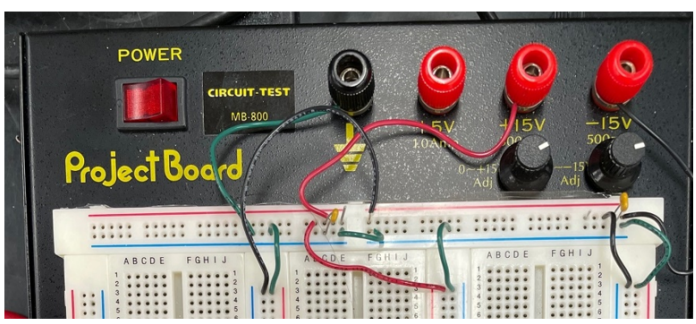
- What Kiers et al. call a "phase portrait" is actually a projection of the motion in phase space. Because the equation is third order, the full phase space is three dimensional, and one should plot the motion in 3d using, e.g., $x$, $\dot{x}$, and $\ddot{x}$. In this full space, trajectories do not cross (why?). In the simpler 2d space of $x$ and $\dot{x}$, trajectories do cross, but this feature is an artifact of the projection. Your interpretation of the motion should take this subtlety into account.
- Estimate the Feigenbaum constant $\delta \approx 4.7$. Hint: If you can measure $\delta_n = \frac{A_n - A_{n-1}}{A_{n+1} - A_n}$ at several levels $n$, try to extrapolate to $n \to \infty$. Here, $A_n$ is the value of $R_v/R$ at which the bifurcation from period $2^n \to 2^{n+1}$ occurs.
- Another interesting quantity to measure is the return map (see Kiers et al. for an example; make sure to be in the chaotic regime!). This is significant because systems with Poincaré sections (return maps, or, more properly, Lorenz maps) that have a single quadratic maximum can all show the period-doubling cascades explored here. Read the discussion in Kiers et al. and Strogatz. Does your return map $x_{n+1}$ vs $x_n$ indicate the presence of an unstable limit cycle? If so, can you show (as Kiers et al. do) an example time series that confirms your conclusion? (If you really are into this point, compute the second-return map $x_{n+2}$ vs $x_n$ and repeat the analysis for a period-2 cycle.)
- Largest Lypapunov exponent. This is the real indicator of chaos but is tricky to estimate. See Strogatz Section 10.5 for hints.
- If you have time, try to estimate $\alpha \approx 2.5$, too. See Strogatz Section 10.6 for a definition

of these two quantities (pp. 379–380). The tricky part is understanding how to define the "width" of the bifurcation branches in a systematic way.

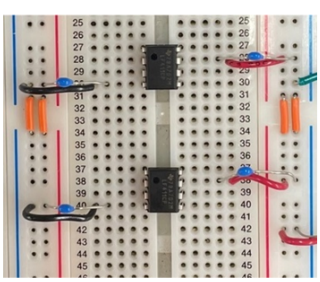
- You might observe further periodic orbits (Period 3, 5, etc.) *beyond* the value of $R_v$ where chaos first appears. The order that these *periodic windows* occur in is also a universal feature of dynamical systems described by return maps such as the one present for this circuit. How many of these periodic windows beyond the primary period-doubling cascade can you find? Note the order (by $R_v$) that they appear in. See Strogatz for more on this *U-Sequence*.

### Breadboarding tips (pch)

- For chaotic circuit use LF411 opamps.
- Use the rails on the breadboard to run +15V, -15V and ground lines across your breadboard - less wiring and cleaner wiring. On some breadboards, the rails do not run the length of the breadboard (check), and jumper as necessary. (See Fig. 4).
- See figure below for tidy powering of opamp with +15 V and -15 V with short jumpers. Note the optional small capacitors (50 or 100 nF) added close to the op-amp from +15V terminal to ground and from -15V terminal to ground to reduce noise into the op-amp. These capacitors direct high frequency noise on the supply lines to ground and away from the op-amp for a quiet circuit.
- For a good circuit keep connecting wires as short as possible. Dangling wires pick up noise and make headaches!
- For verifying and calibrating the nonlinear element's operation, you can do it point by point by hand if you insist but I recommend using a sawtooth ramp input (linear ramp) from the waveform generator and then looking and saving input and output on two scope channels. You see visually immediately if it's working. And for data analysis: one file, all the data saved in a second - boom, done!
- You need a voltage of 0.250-0.500 V for the chaotic circuit. You can build your own adjustable voltage source with a potentiometer to make a resistive voltage divider (Ask TA). Or you can just use the DC power supply provided to create the voltage. If you use the power supply, please connect the power supply's negative terminal (black) to the bread ground to make a common ground reference. The function generator in DC output mode is another option.




*Use breadboard lines to deliver +15V -15V and ground voltages in efficient manner around the breadboard. The two knobs on the right allow you to adjust the voltages to +15 V and -15 V*



*Tidy powering of an op-amp with +15 V (red) and -15 V (black) supply voltages from breadboard rails. Filtering capacitors in blue from +15V line to ground and from -15V line to ground. The orange wires are jumpers to extend the rails across the breadboard.*

**Figure 4:** Breadboard and op-amp powering.